

a SOLVATION PHYSICS

source calculations and response coordinates

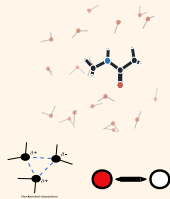
Continuum response

COSMOtherm · SMD(water)



Explicit-water response

OpenFF · interaction coordinates



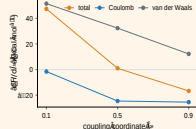
Conformational response

ConfSolv H₂O summaries



Explored high-fidelity response

PIMD2 dH/dλ · not retained



STAGE 1 physical response → frozen structure surrogates

Source calculations are paid for once.

b AMORTIZED RESPONSE

structure-predictable physical priors

15
named scalars

CALCULATED SOLVATION

- 1 COSMOtherm water response
- 9 SMD(water) response

POLARITY / H-BONDING

- 2 Abraham E · excess molar refraction
- 3 Abraham S · dipolarity / polarizability
- 4 Abraham A · H-bond acidity
- 5 Abraham B · H-bond basicity
- 6 Abraham L · hexadecane-air partition

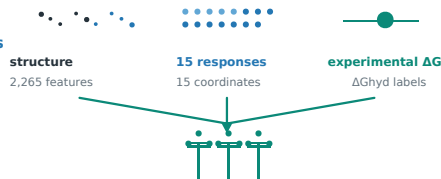
EXPLICIT / IMPLICIT WATER

- 7 corrected OpenFF explicit-water response
- 8 corrected GBn2 implicit-solvent response

CONFORMATIONAL RESPONSE

- 10 gas conformer correction
- 11 solution conformer correction
- 12 hydration conformer correction
- 13 conformer solvation-energy spread
- 14 mean conformer solvent response
- 15 conformer response spread

STAGE 2 experimentally supervised endpoint



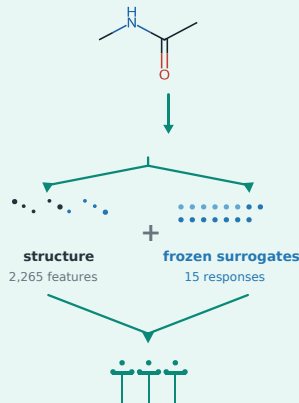
ExtraTrees endpoint ensemble

3 × 360 trees · frozen after evaluation

c DEPLOYMENT

new molecules

NEW MOLECULE



SolvAI

frozen endpoint ensemble

ΔG_{hyd}
kcal mol⁻¹

~~SMILES only · no MD · no PIMD · no probe~~

PIMD8 ≈ 0.205 kcal mol⁻¹

reference only